

# Loop Closure from Two Views: Revisiting PGO for Scalable Trajectory Estimation through Monocular Priors

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**Abstract**—(Visual) Simultaneous Localization and Mapping (SLAM) remains a fundamental challenge in enabling autonomous systems to navigate and understand large-scale environments. Traditional SLAM approaches struggle to balance efficiency and accuracy, particularly in large-scale settings where extensive computational resources are required for scene reconstruction and Bundle Adjustment (BA). However, this scene reconstruction, in the form of sparse pointclouds of visual landmarks, is often only used within the SLAM system because navigation and planning methods require different map representations. In this work, we therefore investigate a more scalable Visual SLAM (VSLAM) approach without reconstruction, mainly based on approaches for two-view loop closures. By restricting the map to a sparse keyframed pose graph without dense geometry representations, our ‘2GO’ system achieves efficient optimization with competitive absolute trajectory accuracy. In particular, we find that recent advancements in image matching and monocular depth priors enable very accurate trajectory optimization without BA. We conduct extensive experiments on diverse datasets, including large-scale scenarios, and provide a detailed analysis of the trade-offs between runtime, accuracy, and map size. Our results demonstrate that this streamlined approach supports real-time performance, scales well in map size and trajectory duration, and effectively broadens the capabilities of VSLAM for long-duration deployments to large environments.

## I. INTRODUCTION

**S**IMULTANEOUS Localization and Mapping (SLAM) methods have become a cornerstone in any autonomous robotic system. A robust, accurate, and scalable SLAM solution is crucial for applications that require navigation over large-scale environments and/or long deployment times. SLAM has been extensively studied over the past few decades [1], [2], with methods deployed across various platforms in diverse environments for tasks such as robot navigation [3]–[5], mixed reality [6], [7], autonomous driving [8], [9], and mobile device applications [10]. SLAM methods have

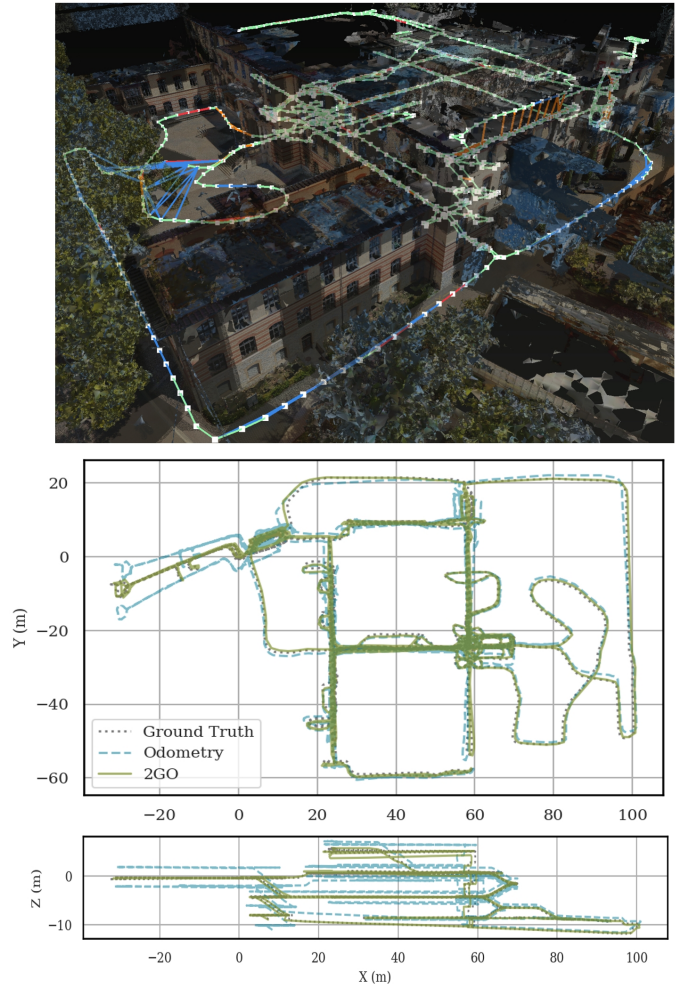


Fig. 1: **Top:** 2GO, our proposed system, refines a 1.1 hr, 2.7 km trajectory collected from a quadrupedal robot, overlaid on a 3D reconstruction of the surroundings. Absolute and scale-free loop-closure edges are visible in yellow and blue, respectively. Loop-closure edges with both types of constraints present are in red. **Bottom:** 2GO significantly improves the quality of the input trajectory, particularly in the Z-direction.

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become remarkably accurate even in very challenging environments [11], but achieving scalability remains challenging due to the substantial computational and storage demands involved. VSLAM methods are of interest due to the widespread availability and low cost of camera sensors, but in competitions such as the HILTI SLAM Challenge [12], their accuracy lags behind LiDAR SLAM. In summary, despite a long line of research, a VSLAM solution that is both scalable and accurate is still missing.

Concerning accuracy, there has been steady progress in

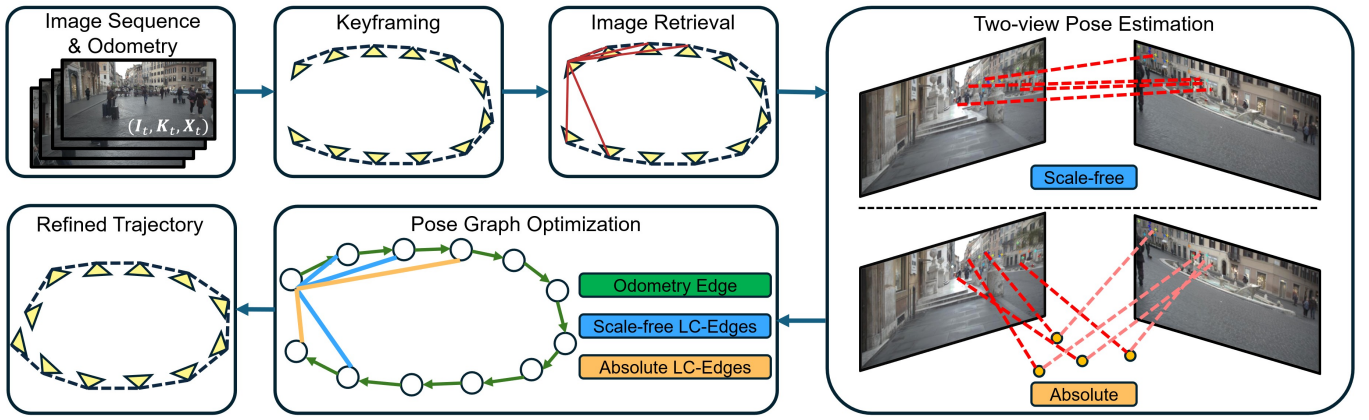


Fig. 2: **System Overview.** From an input stream of odometry and images, we first downsample keyframes. Then we check for loop-closure candidates within previous keyframes and estimate the relative pose between candidate image pairs (metric and/or up-to-scale). Candidates are then filtered using geometric checks and then added as two-view edges to the pose-graph map, which is optimized with PGO. Marked in yellow is the main innovation of this paper.

VSLAM and VIO towards highly accurate pose estimates within a context of about 10min. Stereo visual odometry has even outperformed LiDAR-based methods on the KITTI benchmark [13] (all trajectories in KITTI are shorter than 10min). Locally accurate odometry is available for visual-inertial setups [14], [15], and monocular odometry makes steady progress [16]. However, while visual odometry provides reliable local tracking, achieving globally accurate trajectory estimation requires loop closures and global optimization, where scalability remains challenging. State-of-the-art methods use pose graph optimization (PGO) or bundle adjustment (BA) for a globally consistent trajectory, with BA also providing a consistent reconstruction. PGO has a worst-case time complexity cubic in the number of poses [17], and the time complexity of BA is linear in the number of landmarks and cubic in the number of poses [18]. Consequently, as our experiments confirm, for large environments with many landmarks, and specifically for long trajectories with many poses, the optimization grows in complexity and takes too long to provide real-time updates.

This work investigates reducing SLAM optimization complexity by questioning the necessity of 3D reconstruction. The sparse landmark point cloud typically optimized by BA is rarely used in downstream robotic modules. It is often too sparse for operator visualization, and planning methods typically use other data structures, such as Truncated Signed Distance Fields (TSDF). In practice, fully-fledged mapping frameworks run separate processes to integrate a TSDF from the optimized poses [19]. However, within existing SLAM frameworks, reconstructed landmarks are required to process loop closures and to maintain map consistency over long horizons: Maplab [19] finds loop closures by matching landmarks that are close in descriptor space, applying geometric checks based on the reconstruction, and then merging matched landmarks to the same position. BA then optimizes the poses accordingly. ORB-SLAM3 [15] follows the same approach for loop closure optimization.

In this work, we explore a more scalable approach to VSLAM based on two key insights: (i) robust and **locally accurate odometry** is widely available on most robotic systems and other devices; (ii) large deep-learning models are increasingly adept at estimating geometry from single and multiple views,

providing useful vision priors that enable the construction of **global relations** between keyframes without requiring a map. While the first development reduces the margin of improvement for BA given a trajectory from odometry, the second point prompts us to reconsider if a global geometric map is necessary at all. Thus, we propose to optimize a globally consistent trajectory only through PGO, where loop-closure (LC) edges are constructed directly from pairs of keyframes. To this end, we revisit prior approaches for scale-free LC edges with modern image matchers and experiment with generating absolute LC constraints from metric monocular depth [20] or the MAST3R two-view estimator [21].

Our 2-view PGO approach, which we call ‘2GO’, therefore does not produce any 3D reconstruction. Instead, it uses locally accurate odometry for short-term consistency and vision priors to establish global relations between keyframes, enabling fully map-free optimization using PGO. Through extensive experiments on large-scale trajectories, we show that this simplification yields significantly lower map size and runtime. Surprisingly, in most trajectories, 2GO also yields higher pose accuracy than state-of-the-art VSLAM methods. In summary, our contributions are:

- **2GO**, a two-view PGO framework that leverages the potential of recent advancements in monocular vision priors for computationally efficient and accurate VSLAM which scales to large-scale environments.
- Comprehensive analysis of runtime, accuracy, and map size of VSLAM methods on long trajectories and large scenes.
- Extensive experiments on large-scale datasets, with different odometry sources, and comparisons with a mapping system that performs global BA.

## II. RELATED WORK

### A. Large-Scale SLAM

Conventional VSLAM and VIO approaches [14], [15] build a metric 3D map and jointly optimize the map and camera pose. They have demonstrated reliable performance across various public benchmarks [12], [22], [23]. These methods achieve high local accuracy, but often face scalability challenges. To improve the scalability of VSLAM, previous methods have explored reducing the computational cost of

BA by limiting it to a sliding window or running it at a low frequency [15], [24]. Some approaches eliminate BA entirely and rely solely on PGO, though they still require landmarks to establish loop closure constraints [14]. In contrast, our approach not only skips BA but also eliminates the reconstruction and triangulation steps.

### B. Vision-based Priors

Recent SLAM methods have explored monocular depth to enhance accuracy and robustness. For instance, recent Neural-SLAM methods, including [25], leverage monocular depth estimators for dense reconstruction. Predicting geometric priors from monocular inputs, such as depth and surface normals [20], has made significant progress. However, these monocular priors remain underexplored in SLAM, particularly in sparse monocular settings. It remains unclear how to effectively use depth priors without performing dense reconstruction.

Beyond advances in monocular priors, two-view tasks have also seen significant progress, including feature matching [26], visual place recognition (VPR) [27], and two-view reconstruction [21]. Improvements in feature detectors and matchers have enabled robust 2D-2D matching in challenging scenarios, such as wide-baseline settings. Global VPR methods have been widely explored and deployed in visual localization [28], with increasing attention on map-free localization due to their efficiency on edge devices, such as those used in augmented reality (AR) [29]. For instance, *MARLoc* [30] integrates deep-learned 2D features and matchers with global image descriptors, achieving precise map-free localization on AR devices. A similar concept can be applied to SLAM to enhance loop-closure edge construction in a map-free manner. Two-view reconstruction priors have also been initially explored in both Structure-from-Motion (SfM) [31] and VSLAM [32]. However, these methods still rely on reconstruction maps and perform dense bundle adjustment.

## III. LOOP CLOSURES FROM TWO VIEWS

Figure 2 provides an overview of 2GO. To constrain a trajectory using loop closures (LCs) without reconstructing 3D geometry, 2GO first finds LC candidates via image retrieval from previous keyframes and then employs different pose constraints between image pairs. Finally, pose graph optimization (PGO) is performed using the identified LC-Edges to refine the prior trajectory derived from odometry. In this section we introduce how we construct different variants of loop-closure edges from two views. In Section IV we describe how these two-view constraints are used in the proposed SLAM system.

### A. Problem Statement

Given a sequence of images from a calibrated camera and their corresponding odometric poses, the input to 2GO at time  $t$  is  $(\mathcal{I}_t, \mathbf{X}_t)$ .  $\mathcal{I}_t \in \mathbb{R}^{H \times W \times C}$  is the rectified image from the camera, and  $\mathbf{X}_t \in \text{SE}(3)$  is the pose of the camera with respect to a reference frame. 2GO incrementally refines the overall trajectory up to the present timestamp  $N$ ,  $\hat{\mathbf{X}} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \dots, \hat{\mathbf{X}}_N)$ . Without loss of generality, we describe the following methodology for a single-camera system. However, the approach extends naturally to multiple cameras, provided that the inter-camera extrinsics are known.

### B. Image Retrieval

The image retrieval stage identifies LC candidates—image pairs that can establish two-view constraints by observing the same region of the environment. Given a new input keyframe  $\mathcal{I}_k$ , image retrieval searches previous keyframes  $\mathcal{I}_{1,2,\dots,k-1}$  for LC candidates, according to their *appearance* and *geometric* similarity, analogous to the *long-term* and *mid-term* data associations described in [15]. Unlike conventional image retrieval methods for LC detection, 2GO prioritizes high recall over high precision, as multiple filtering stages (detailed in Section IV-C), mitigate outliers introduced during image retrieval. This aggressive strategy increases the likelihood of detecting valid LCs to refine the input trajectory.

**Similarity Candidates:** While many SLAM systems utilize bag-of-words based VPR [33], only a few SLAM systems use global-descriptor based VPR approaches [34]. These are more commonly used in SfM methods [28], where image correspondences are retrieved from large collections of unordered images. 2GO leverages one such VPR model, *BoQ* [35], which generates a global descriptor  $\mathbf{F}_t$  for each image  $\mathcal{I}_t$ , to identify image pairs with visual similarity.

The VPR model is trained to ensure that the descriptors of two images are close in a similarity metric  $\langle \mathbf{F}_a, \mathbf{F}_b \rangle$  (in our case the dot product), if  $\mathcal{I}_a$  and  $\mathcal{I}_b$  share visual overlap. Hence, we define a set of similarity LCs,  $\mathcal{C}_{\text{sim}}$  for a given image  $\mathcal{I}_k$  by selecting the  $n_{\text{sim}}$  images with most similar descriptors:

$$\mathcal{C}_{k,\text{sim}} = \left\{ \mathcal{I}_j \mid j \in \text{Top}_{j=0\dots k-1}^{n_{\text{sim}}} \langle \mathbf{F}_k, \mathbf{F}_j \rangle \right\} \quad (1)$$

where  $\text{Top}_A^n(\cdot)$  picks the top  $n$  elements out of  $A$  according to the given metric.

**Proximity Candidates:** We also consider candidate LCs based on the geometric difference between the current and previous keyframe poses. This allows retrieving images that might otherwise be missed due to low visual similarity, and hence have largely different global descriptors, such as those affected by large viewpoint changes. Since the robust learned matchers [21], [36] used in the subsequent steps can establish correspondences even across wide baselines, incorporating such LC candidates is beneficial, as seen in Figure 5. To accomplish this, we first find the set of nearby previous keyframes:

$$\tilde{\mathcal{C}}_{k,\text{prox}} = \left\{ \mathcal{I}_j \mid \forall j : \text{Rot}(\hat{\mathbf{T}}_j^k) < \delta_r \wedge \text{Trans}(\hat{\mathbf{T}}_j^k) < \delta_t \right\} \quad (2)$$

where  $\hat{\mathbf{T}}_j^k = \hat{\mathbf{X}}_k^{-1} \hat{\mathbf{X}}_j$  denotes the relative transform between the current estimate of the poses of keyframes  $i$  and  $j$ , and  $\delta_r$  and  $\delta_t$  are the thresholds for rotational and translational differences, respectively. We prioritize matches with a larger time gap  $\Delta t(\mathcal{I}_k, \mathcal{I}_j)$  between images, as they are more effective in correcting odometry drift:

$$\mathcal{C}_{k,\text{prox}} = \text{Top}_{\mathcal{I}_j \in \tilde{\mathcal{C}}_{k,\text{prox}}}^{n_{\text{prox}}} \Delta t(\mathcal{I}_k, \mathcal{I}_j) \quad (3)$$

Subsequently, LC candidates  $\mathcal{C} = \mathcal{C}_{\text{sim}} \cup \mathcal{C}_{\text{prox}}$  are passed to the two-view pose estimation module.

### C. Constructing Two-View LC-Edges

In conventional VSLAM and SfM methods, image poses are typically estimated via PnP using 2D-3D correspondences

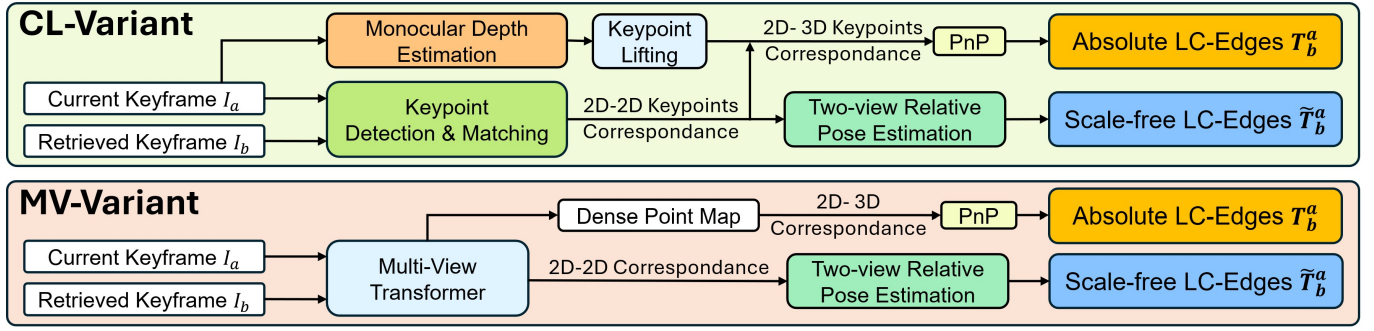


Fig. 3: **Two-view Pose Estimation.** Both CL and MV variants establish 2D-2D and 2D-3D feature correspondences for scale-free and absolute LC-Edges, respectively. The CL-variant employs separate deep learning models in a modular pipeline, while the MV-variant achieves this with a single two-view reconstruction model.

from an estimated metric map. Early Pose-SLAM approaches also relied on landmarks to define LC constraints without optimizing their positions [37]. 2GO eliminates the need for explicit 3D landmarks by (i) revisiting up-to-scale LC-edges from 2D-2D matches and (ii) leveraging metric monocular depth prior for single-view 3D geometry estimation.

For each keyframe  $I_k$ , 2GO estimates its relative pose  $T_a^k \in \text{SE}(3)$  to each LC candidate  $I_a \in \mathcal{C}$ . Figure 3 illustrates two variants of this process for obtaining *scale-free*  $\tilde{T}_a^k$  and *absolute*  $T_a^k$  two-view LC-edges. The **CL** (Correspond-Lift) variant follows conventional pipelines, using separate models for keypoint detection [38], matching [36], and keypoint depth estimation via metric monocular depth (MDE) [20]. In contrast, the **MV** (Multi-View) variant employs a multi-view model such as MAST3R [21], which predicts 2D keypoint matches and 2D-3D correspondences through a single model.

**Scale-free LC-Edge:** The 2D keypoint correspondences between two images yield the essential matrix  $\mathbf{E}$  [39]. The up-to-scale relative pose between the two keyframes  $\tilde{T}_a^k = [\mathbf{R}_a^k | \tilde{\mathbf{t}}_a^k]$  can then be obtained by decomposing  $\mathbf{E} = [\tilde{\mathbf{t}}_a^k]_{\times} (\mathbf{R}_a^k)^T$ . Similar as [40], the scale-free pose residual  $\mathbf{r}_S(\tilde{T}_b^a, \tilde{T}_b^a)$  is computed as the pose error between a scale-free pose measurement  $\tilde{T}_b^a$  and the expected measurement  $\hat{T}_b^a$  from the current estimate in the pose graph.

$$\mathbf{r}_S(\tilde{T}_b^a, \tilde{T}_b^a) = \begin{bmatrix} \tilde{\mathbf{t}}_b^a - \text{norm}(\hat{\mathbf{t}}_b^a) \\ \mathbf{R}_b^a \boxminus \hat{\mathbf{R}}_b^a \end{bmatrix} \in \mathbb{R}^6 \quad (4)$$

$$\text{norm}(\mathbf{t}) = \begin{cases} \frac{\mathbf{t}}{\|\mathbf{t}\|}, & \|\mathbf{t}\| > 0, \\ \mathbf{0}, & \|\mathbf{t}\| = 0. \end{cases} \quad (5)$$

Here,  $\boxminus$  denotes the operation mapping the difference in expected and measured rotations from the Lie Group  $\text{SO}(3)$  to a vector in  $\mathbb{R}^3$  for optimization [41]. In a scale-free pose measurement, the translation vector  $\tilde{\mathbf{t}}_b^a$  is a unit vector. The expected translation  $\hat{\mathbf{t}}_b^a$  is therefore also normalized before the two translation vectors are compared. The rare case  $\|\tilde{\mathbf{t}}\| = 0$  is a pure rotation. To prevent numerical instabilities, these residuals are not included in the pose graph.

While one scale-free LC-Edge does not constrain the scale of the pose between two keyframes, they can still constrain the pose graph if enough scale-free edges are added. This is demonstrated in our experimental evaluations.

**Absolute LC-Edge:** In the CL-variant, a MDE model is used to obtain a metric depth map  $\mathbf{D}_k \in \mathbb{R}^{H \times W}$  from  $I_k$ .  $\mathbf{D}_k$

is used to lift the detected keypoints  $\mathbf{p}_0^k, \dots, \mathbf{p}_N^k$  to 3D.

$$\mathbf{P}_i^k = d_i^k \mathbf{K}^{-1} \begin{bmatrix} \mathbf{p}_i^k \\ 1 \end{bmatrix} \quad (6)$$

Here,  $d_i^k$  is the estimated depth value at the pixel  $\mathbf{p}_i^k$ , and  $\mathbf{K}$  is the camera intrinsic matrix.

In the MV-variant, a model predicts a dense point-map  $\mathbf{D}' \in \mathbb{R}^{H \times W \times 3}$ , where each pixel in  $I_k$  has a predicted position relative to  $\mathbf{X}_k$ . The coordinates of  $\mathbf{D}'$  indexed at  $(u, v) = \mathbf{p}_i^k$ , directly give  $\mathbf{P}_i^k$ .

Both two variants generate 3D-2D correspondences  $(\mathbf{P}_i^k, \mathbf{p}_i^a)$  between  $I_k$  and  $I_a$ . These allow a PnP solver to obtain an absolute pose between them, yielding  $T_a^k$ , which is incorporated into the pose graph as an absolute LC-Edge. Some MV-Models can alternatively directly estimate  $T_a^k$  [42].

#### IV. SYSTEM IMPLEMENTATION

This section introduces how we build a complete trajectory estimation system around the two-view LC mechanism introduced in Section III. Given the raw input image and odometry streams, we subsample the camera stream into keyframes. During the image retrieval as introduced in Section III-B, we run filtering steps to rule out potential false-positive LCs. This loop is repeated for every new incoming keyframe and we construct the PGO that includes both filtered scale-free and absolute LC-edges to regularly optimize the trajectory.

##### A. Keyframing

2GO enforces spatial sparsity by choosing new keyframes solely based on the translational or rotational distance from the current frame to the previous keyframe. 2GO does not keep a landmark map, making landmark triangulation between keyframes unnecessary. Hence, consecutive keyframes do not require a minimum number of covisible keypoints, allowing for a sparser selection strategy that enhances scalability.

##### B. Two-View Pose Estimation

To compute relative pose between two LC frames constructed in Section III-C, 2GO uses the *COLMAP* [43] implementation of relative pose estimation. For keypoint correspondences between images, the CL-variant pipeline uses *DISK* features [38] and the *LightGlue* [36] matcher. We then estimate the metric depth through *Metric3Dv2* [20]. For the MV-Variant, we use *MASt3R* [21]. 2GO is not tightly coupled to using these choices. As newer models with better performance become available, they can be swapped in easily.

### C. LC Filtering

Before integration into the pose graph, LC-Edges undergo feasibility filtering. This enhances robustness against false positives in image retrieval and minimizes redundant computation, in turn reducing overall runtime.

**Image Matching Feasibility:** Scale-free and absolute LC-Edges require sufficient 2D-2D and 2D-3D correspondences, respectively. This filter uses the image matchers to ensure that images have enough meaningful visual overlap.

**Geometric Feasibility:** During computation of scale-free and absolute LC-Edges, the respective solvers assess if the correspondences yield a consistent model, ensuring the geometric consistency of the correspondences from image matching.

**Pose-Graph Consistency:** The computed transform between images should remain consistent with the current optimized trajectory. This enhances robustness against false-positive matches in visually similar environments.

### D. Pose Graph Optimization

Nonlinear least-squares is used to optimize the pose graph. The *iSAM2* incremental optimizer implemented in *GTSAM* [17] is used as the back-end for optimization. In 2GO, the objective function of its PGO is formulated as:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \underbrace{\|\mathbf{r}_P(\mathbf{X}_0)\|_{\Sigma_P}^2}_{\text{prior}} + \underbrace{\sum_{e \in \mathcal{O}} \|\mathbf{r}_A(h(\mathcal{X}), e)\|_{\Sigma_e}^2}_{\text{odometry edges}} + \underbrace{\sum_{l \in \mathcal{L}_a} \rho(\|\mathbf{r}_A(h(\mathcal{X}), l)\|_{\Sigma_{l_a}}^2)}_{\text{absolute LCs}} + \underbrace{\sum_{l \in \mathcal{L}_s} \rho(\|\mathbf{r}_S(h(\mathcal{X}), l)\|_{\Sigma_{l_s}}^2)}_{\text{scale-free LCs}} \quad (7)$$

where  $\|\mathbf{a}\|_{\Sigma}^2 = \mathbf{a}^T \Sigma \mathbf{a}$  denotes the L2 norm of  $\mathbf{a} \in \mathbb{R}^n$  weighted by  $\Sigma \in \mathbb{R}^{n \times n}$ .  $h(\mathcal{X})$  is the measurement function that obtains the relevant expected measurement from  $\mathcal{X}$ .  $\rho(\cdot)$  denotes a robust Cauchy loss [44]. The overall error sums over odometry edges, absolute LC-Edges, and scale-free LC-Edges. Three types of residual are defined:

- $\mathbf{r}_A(\hat{\mathbf{T}}_b^a, \mathbf{T}_b^a) \in \mathbb{R}^6$ , representing the pose error between an absolute pose measurement  $\mathbf{T}_b^a \in \text{SE}(3)$  from frame  $b$  to  $a$  and its expected measurement  $\hat{\mathbf{T}}_b^a \in \text{SE}(3)$  in the pose graph. This residual is well-studied and commonly used in PGO applications [14]. The difference in expected and measured poses is mapped to a vector space  $\mathbb{R}^6$  from  $\text{SE}(3)$  for optimization [41].
- $\mathbf{r}_P(\mathbf{X}_0) = \mathbf{r}_A(\hat{\mathbf{X}}_0, \mathbf{X}_0) \in \mathbb{R}^6$  is a prior fixing the current estimate of the first pose  $\hat{\mathbf{X}}_0$  to its initial value  $\mathbf{X}_0$ .
- $\mathbf{r}_S(\hat{\mathbf{T}}_b^a, \tilde{\mathbf{T}}_b^a) \in \mathbb{R}^6$ , is the pose error between a scale-free pose measurement  $\tilde{\mathbf{T}}_b^a$  and the expected measurement  $\hat{\mathbf{T}}_b^a$  from the current estimate in the pose graph, as Equation (4) introduced in Section III-C.

PGO is performed every  $\lfloor N/l \rfloor$  keyframes, where  $N$  is the total number of keyframes and  $l$  is a tunable parameter. This keeps optimization tractable as the pose graph grows denser.

## V. EXPERIMENTS

### A. Experiment Setup

2GO focuses on optimizing a trajectory from an odometry source in large-scale scenarios. Among common SLAM

datasets, like EuRoC [23], and TUM-VI [45], KITTI offers the longest trajectories with dense ground truth, making it a representative conventional benchmark. In addition, we select Vision Benchmark in Rome (VBR) [46], a recent dataset with long-duration, large-scale sequences. To further demonstrate 2GO's real-world applicability, we evaluate it on a large-scale robot recording spanning 1.1 hr and 2.7 km.

We use two VSLAM methods as baselines. **VINS-Fusion** [47] and **ORB-SLAM3** [15] are both sliding-window methods with bag-of-words place recognition. When a loop closure is detected, VINS-Fusion performs 4-DOF PGO, while ORB-SLAM3 performs BA. Both methods are run in stereo mode without IMU input. KITTI has no high-rate IMU data, and omitting IMU for VBR yielded better trajectories.

In addition, we compare 2GO to **Maplab2.0** [19], a mapping framework capable of fusing multimodal sensor inputs. It can take an external odometry source as input and optimize its trajectory based on LCs and BA. It uses covisible visual keypoints for place recognition and performs loop closure by merging covisible keypoints between keyframes and performing global BA. We evaluate variants using BRISK [48] (**BIN**) and SuperPoint [49] (**SP**) features. Unless otherwise stated, the CL-variant of 2GO is used, which is justified in Section V-C.

All methods are run in an offline single-threaded batch-processing mode, where sensor inputs are read from file and processed as fast as possible. This measures the total runtime of all components of each algorithm.

### B. Is 2GO more scalable?

We first show 2GO's runtime and storage scalability in large-scale scenarios. Table II shows the runtime and storage size of different methods on the longest handheld VBR sequence, *spagna\_train0*, with a distance of 1.56 km and a 23 min 34 sec duration. 2GO only requires  $0.83 \times$  of the real-time duration to process the sequence. The only other method within real-time is Maplab-BIN, while ORB-SLAM3 and VINS-Fusion have total processing time close to real time.

Two baselines are much slower: VINS-Fusion performing PGO takes  $4.4 \times$  real-time to perform global PGO, as it uses the same keyframing strategy for odometry and pose refinement, resulting in more than 10000 keyframes. In contrast, 2GO optimizes 740 keyframes. Though both methods do not perform BA, this illustrates the growing time complexity of PGO with the number of input poses. The second outlier, Maplab-SP, requires about  $10 \times$  real-time to generate a map of camera poses and landmarks, as it uses the computationally expensive SuperPoint and SuperGlue models to detect and match keypoints between all consecutive frames.

Conversely, Maplab-BIN uses lightweight BRISK features, completing its mapping stage in  $0.08 \times$  real-time. However, these features are less discriminative and less robust. As fewer loop closures are found, the optimization duration is shorter. Therefore, it cannot reduce the input odometry's ATE.

We compare map sizes between Maplab, ORB-SLAM3 and 2GO, as VINS-Fusion lacks map-saving functionality. The maps saved by Maplab and ORB-SLAM3 minimally include the pose graph, as well as the landmark locations, descriptors, and keypoint coordinates in each keyframe. In contrast, 2GO

| Sequence                   | 2GO Variant   | Median Trans. Err. | Median Rot. Err. | #LC-Edges | ATE (m) ↓ | % Decrease ↑ | Runtime (sec) ↓ | % real-time ↓ |
|----------------------------|---------------|--------------------|------------------|-----------|-----------|--------------|-----------------|---------------|
| VBR <i>ciampino_train0</i> | CL Scale-free | 0.87 deg           | 0.37 deg         | 6200      | 23.36     | <b>82.20</b> | 972.51          | 62.83         |
|                            | CL Absolute   | 0.25 m             | 0.49 deg         | 3403      | 25.12     | 80.86        | 1,986.80        | 128.35        |
|                            | CL Both       | /                  | /                | 10180     | 24.82     | 81.09        | 2,079.34        | 134.33        |
|                            | MV Scale-free | 0.86 deg           | 0.35 deg         | 2933      | 25.58     | 80.51        | 1,705.63        | 110.19        |
|                            | MV Absolute   | 2.12 m             | 2.56 deg         | 3349      | 24.65     | 81.22        | 1,713.51        | 110.70        |
|                            | MV Both       | /                  | /                | 6686      | 24.87     | 81.05        | 1,759.78        | 113.69        |
| KITTI-00                   | CL Scale-free | 3.02 deg           | 0.40 deg         | 1230      | 1.93      | 85.23        | 187.93          | 40.04         |
|                            | CL Absolute   | 0.86 m             | 0.64 deg         | 517       | 1.47      | <b>88.74</b> | 365.08          | 77.78         |
|                            | CL Both       | /                  | /                | 1735      | 1.61      | 87.72        | 365.45          | 77.85         |
|                            | MV Scale-free | 4.82 deg           | 0.49 deg         | 322       | 1.83      | 85.99        | 215.25          | 45.86         |
|                            | MV Absolute   | 4.82 m             | 1.02 deg         | 360       | 2.25      | 82.83        | 210.95          | 44.94         |
|                            | MV Both       | /                  | /                | 698       | 1.77      | 86.49        | 214.77          | 45.75         |

TABLE I: **Performance comparison of different variants and LC-Edge combinations of 2GO.** VINS-Fusion odometry is used for both sequences. The best performing 2GO variant for each dataset is **bolded**. For *ciampino\_train0*, RMSE ATE: 131.26 m, real-time duration: 1547.9 sec. For KITTI-00, RMSE ATE: 13.09 m, real-time duration: 469.4 sec.

| Variant         | VINS-Fusion | VINS-PGO | ORB-SLAM3 | Maplab-BIN                          | Maplab-SP                | 2GO (Ours)  |
|-----------------|-------------|----------|-----------|-------------------------------------|--------------------------|-------------|
| Runtime (sec) ↓ | 1564        | 6206     | 1755      | <b>110</b> (Map)<br><b>29</b> (Opt) | 14139 (Map)<br>271 (Opt) | 1167        |
| Map Size (MB) ↓ | /           | /        | 1136      | 319                                 | 572                      | <b>146</b>  |
| RMSE ATE (m) ↓  | 21.14       | 13.22    | 5.62      | 20.83                               | 20.35                    | <b>3.73</b> |

TABLE II: **Comparison of runtime and map size.** Result across different methods for the *spagna\_train0* trajectory of VBR (1414s real-time duration). “VINS-PGO” denotes VINS-Fusion with loop closure enabled. “Maplab-BIN” and “Maplab-SP” denote Maplab pipeline using Binary feature with nearest neighbor matcher and SuperPoint feature with SuperGlue matcher.

saves a pose graph, as well as a compressed image and a global descriptor for each keyframe. 2GO needs less than half of the storage of Maplab-BIN, a quarter of Maplab-SP, and a tenth of ORB-SLAM3. However, none of the maps are prohibitively large considering the trajectory duration.

This shows that 2GO scales well to long trajectories, even while yielding the lowest trajectory error. It remains one of the few methods within real-time constraints while maintaining the smallest map representation among all baselines.

### C. How do different LC-Edges contribute to the optimization?

We assess 2GO’s performance across CL/MV variants and different LC-Edge combinations. Table I shows results on the KITTI-00 and *ciampino\_train0* sequences with VINS-Fusion odometry. LC-Edge accuracy is measured by comparing the estimated transformation  $\mathbf{T}_a^k$  between two keyframes  $k$  and  $a$  with the transform between their ground-truth poses,  $\mathbf{X}_k^{\star^{-1}}\mathbf{X}_a^{\star}$ . The translation error for Scale-Free LC-Edges is the angle between the normalized unit vectors  $\tilde{\mathbf{t}}_a^k$  and  $\text{norm}(\mathbf{t}_a^k)$ .

CL and MV variants using Absolute and Scale-Free (“Both”) LC-Edges perform well on both sequences. Scale-Free LC-Edge accuracy is similar across both variants, while Absolute LC-Edges from CL are much more accurate. The CL variant finds more LC-Edges. In addition, Scale-Free LC-Edges can refine trajectories, with both variants using only these LC-Edges reducing the ATE except for MV with KITTI-00, due to the low number of LC-Edges found.

In terms of runtime, where Absolute LC-Edges are used, MV variants run faster, perhaps because MAST3R performs both keypoint matching and depth estimation jointly. However, when only Scale-Free LC-Edges are used, the CL does not require MDE, and thus enjoys much lower runtime.

Overall, because the CL Both variant finds many accurate LC-Edges, it is likely to be more robust across varying scenes, and thus is used for all subsequent experiments.

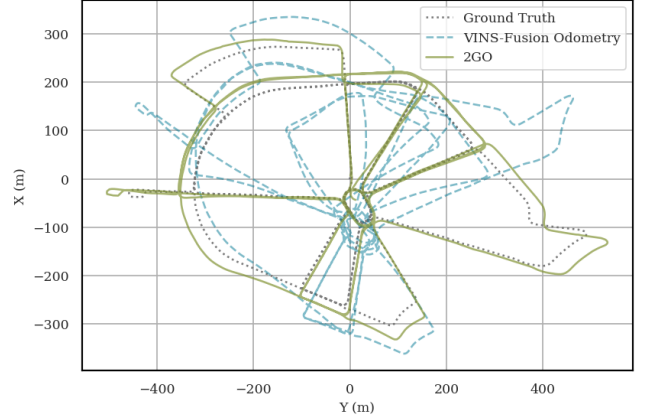


Fig. 4: *ciampino\_train0* sequence: 2GO improves on VINS-Fusion odometry, reducing ATE from 131.33 m to 22.84 m.

### D. How accurate is 2-View PGO?

We evaluate global trajectory error by comparing the RMSE Absolute Trajectory Error (ATE), after applying a 6-DOF alignment between the input and ground-truth trajectories. Results are shown in Table III. To ensure fair evaluation of Maplab and 2GO as trajectory refinement methods, the loop-closing functionality of ORB-SLAM3 is disabled when used as odometry. In summary, 2GO improves upon input odometry in most cases and often achieves the highest accuracy among the compared PGO and BA-based methods.

On the long trajectories in VBR, 2GO strongly outperforms VINS-PGO and Maplab in refining the same input odometry, showing the effectiveness of integrating learned priors in PGO. While the performance gap between 2GO and ORB-SLAM3 performing BA is smaller, long, loopy sequences like *ciampino\_train0* emphasize 2GO’s ability to capitalize on loop closures for trajectory refinement. Several sequences in KITTI, for instance KITTI-01, a highway scene, do not contain loops, so 2GO cannot improve upon input odometry as significantly.

2GO assumes locally continuous and smooth odometry. For instance, when ORB-SLAM3 lost visual tracking in the *diag\_train0* sequence, resulting in discontinuous, jagged odometry, 2GO could not refine it. However, note that all other baselines compared similarly failed.

Figure 4 illustrates 2GO refining VINS-Fusion odometry on *ciampino\_train0*, reducing ATE from 131.33 m to 22.84 m. Figure 5 shows an example of 2GO constructing and leveraging loop-closure edges from image pairs with significantly

| Sequence        | Length  | Duration | w/ ORB-SLAM3 Odometry |             |            |           |              | w/ VINS-Fusion Odometry |             |            |           |              |
|-----------------|---------|----------|-----------------------|-------------|------------|-----------|--------------|-------------------------|-------------|------------|-----------|--------------|
|                 |         |          | odom only             | ORB-SLAM3   | Maplab-BIN | Maplab-SP | 2GO          | odom only               | VINS-PGO    | Maplab-BIN | Maplab-SP | 2GO          |
| KITTI-00        | 3.72 km | 470 sec  | 4.20                  | 4.85        | 37.51      | 12.69     | <b>1.82</b>  | 13.09                   | 4.20        | 11.13      | 11.39     | <b>1.58</b>  |
| KITTI-01        | 2.45 km | 114 sec  | 12.71                 | 13.62       | 116.91     | 57.40     | <b>11.00</b> | 7.95                    | 7.98        | 66.31      | 20.92     | <b>7.93</b>  |
| KITTI-02        | 5.07 km | 483 sec  | 8.05                  | <b>5.64</b> | 18.01      | 44.78     | 8.04         | 21.02                   | 13.03       | 23.96      | 18.29     | <b>10.58</b> |
| KITTI-03        | 0.56 km | 83 sec   | 1.29                  | 1.38        | 4.31       | 5.68      | <b>1.12</b>  | 1.64                    | 1.55        | 4.04       | 4.31      | <b>1.17</b>  |
| KITTI-04        | 0.39 km | 28 sec   | <b>0.24</b>           | 0.34        | 0.93       | 2.65      | 0.28         | 1.30                    | <b>1.25</b> | 2.27       | 2.47      | 1.31         |
| KITTI-05        | 2.21 km | 288 sec  | 2.05                  | 0.95        | 7.44       | 7.98      | <b>0.92</b>  | 6.39                    | 3.68        | 11.37      | 11.21     | <b>1.70</b>  |
| KITTI-06        | 1.23 km | 114 sec  | 2.05                  | 0.79        | 7.09       | 10.74     | <b>0.62</b>  | 3.54                    | 1.68        | 13.27      | 14.76     | <b>1.48</b>  |
| KITTI-07        | 0.69 km | 114 sec  | 1.19                  | 0.50        | 5.60       | 5.63      | <b>0.41</b>  | 2.11                    | 0.65        | 7.48       | 7.70      | <b>0.46</b>  |
| KITTI-08        | 3.22 km | 423 sec  | <b>3.50</b>           | 3.67        | 40.79      | 11.65     | 3.78         | 10.43                   | 10.01       | 17.04      | 14.27     | <b>4.03</b>  |
| KITTI-09        | 1.71 km | 165 sec  | 2.87                  | 3.22        | 10.23      | 11.32     | <b>1.84</b>  | 7.80                    | 7.71        | 15.97      | 12.31     | <b>4.66</b>  |
| KITTI-10        | 0.92 km | 125 sec  | 1.31                  | 1.19        | 44.09      | 11.05     | <b>0.91</b>  | 3.77                    | 3.73        | 9.17       | 9.46      | <b>1.57</b>  |
| campus_train0   | 2.73 km | 602 sec  | 28.92                 | 11.94       | 30.73      | 28.22     | <b>9.32</b>  | 47.74                   | 43.84       | 49.46      | 46.28     | <b>12.30</b> |
| campus_train1   | 2.95 km | 584 sec  | 28.56                 | <b>7.99</b> | 30.88      | 27.42     | 8.98         | 46.94                   | 23.40       | 44.43      | 34.93     | <b>9.60</b>  |
| ciampino_train0 | 9.01 km | 1549 sec | 82.82                 | 68.11       | 83.89      | 81.77     | <b>18.29</b> | 131.33                  | 95.95       | 130.20     | 130.94    | <b>22.84</b> |
| ciampino_train1 | 5.20 km | 942 sec  | 25.98                 | 13.83       | 27.61      | 25.00     | <b>13.25</b> | 47.21                   | 34.68       | 40.94      | 40.79     | <b>14.20</b> |
| colosseo_train0 | 1.45 km | 882 sec  | 11.09                 | 9.67        | 9.40       | 10.29     | <b>6.80</b>  | 21.49                   | 13.37       | 21.48      | 21.47     | <b>6.00</b>  |
| diag_train0     | 1.04 km | 1003 sec | /                     | /           | /          | /         | /            | 27.35                   | 33.10       | 27.17      | 27.23     | <b>26.70</b> |
| pincio_train0   | 1.28 km | 1114 sec | 6.30                  | <b>2.08</b> | 6.31       | 6.91      | 2.79         | 11.29                   | 7.75        | 11.22      | 11.23     | <b>3.18</b>  |
| spagna_train0   | 1.56 km | 1414 sec | 11.42                 | <b>5.62</b> | 11.95      | 12.01     | 6.48         | 21.14                   | 13.22       | 20.83      | 20.35     | <b>3.73</b>  |

TABLE III: RMSE ATE (m) after 6-DOF trajectory alignment. With ORB-SLAM3 odometry, “odom only” is ORB-SLAM3 without loop closures, while “ORB-SLAM3” has loop closures enabled. “VINS-PGO” is VINS-Fusion with loop closure enabled. “Maplab-BIN” is Maplab using Binary features with nearest neighbor matcher. “Maplab-SP” is Maplab using SuperPoint features with SuperGlue matcher. The best method for each odometry source is **bolded**. ORB-SLAM3 lost visual tracking on the *diag\_train0* VBR sequence.

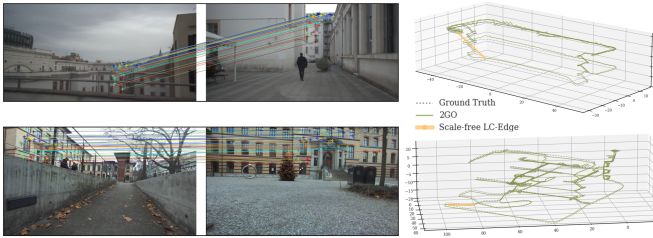


Fig. 5: **Scale-free LC Edge Examples.** The top row contains samples from *diag\_train0*, while the bottom row is from our real-world recording. Both LCs are constructed from image pairs with large baselines and contribute to trajectory refinement.

|              | Spot Odom | ZED Odom | ZED w/ LC | 2GO         |
|--------------|-----------|----------|-----------|-------------|
| RMSE ATE (m) | 2.44      | 26.18    | 23.05     | <b>0.66</b> |

TABLE IV: **Comparison of methods on a real-world trajectory.** Spot Odom is used as odometry input for 2GO. Over a 1.1 hr real-world trajectory, 2GO has cumulative runtime of only 15.0 min, and saves a 162.5 MB map.

different viewpoints.

#### E. How does 2GO work on a real-world robot?

We assess the accuracy, scalability and real-world applicability of 2GO with a 1.1 hr, 2.7 km data recording on a Boston Dynamics Spot robot. The trajectory primarily covers different areas of a multi-floor building and a short outdoor segment, as shown in Figure 1. A Stereolabs Zed 2 camera provides images, IMU, and on-device VIO. Spot also provides device odometry, fused from leg and visual sources.

We use the pipeline of [50] to generate ground truth poses. Our results in Table IV confirm that 2GO can be readily deployed on existing platforms using their provided odometry. It consistently operates within real-time constraints while maintaining a low-memory footprint for the map.

### VI. CONCLUSION

We presented 2GO, a novel approach to scalable visual SLAM that removes the need for explicit scene reconstruction by leveraging two-view loop closures and monocular priors. Constraining the map to a sparse keyframed pose graph yields

an efficient optimization process while maintaining competitive trajectory accuracy. Our experiments on diverse datasets, including large-scale scenarios, highlight the effectiveness of recent advancements in image matching and monocular metric depth estimation for accurate trajectory estimation. The results show that 2GO enables real-time performance, scales effectively in both map size and trajectory duration, and is well-suited for long-duration deployments in large environments. This work broadens the scope of visual SLAM by offering a practical alternative for scenarios where dense reconstruction is unnecessary, paving the way for efficient and scalable solutions in autonomous navigation and mapping.

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